

Parameter-disturbance robustness of the exact DCT-IV init

Controlled O(2)-twiddle DCT-IV on DIV2K-8q: Gaussian jitter ($\sigma = 0.1$) of a random fraction of the 2200 exact-init gate entries, 1008-step top-10% training, 3 seeds

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Setup

Starting from the **exact** analytic DCT-IV (DCT4Basis(8, 8, parametrization: "controlled") — the O(2)-twiddle form of pdfT PR #24, commit 5365a5a, whose CRY twiddle is applied to the control = 1 half only), we perturb a random fraction f of its **2200 real gate-tensor entries** with on-manifold Gaussian jitter: add $N(0, 0.1)$ to the selected entries and re-project each touched gate onto its manifold (nearest-orthogonal O(4)/O(2) via SVD; the Δ -sign gate by a phase jitter), so the perturbed init stays a valid real-orthogonal DCT-IV and its untrained PSNR is interpretable. The disturbed counts are $\lfloor f \cdot 2200 \rfloor = \{2, 4, 11, 22, 44, 110, 220, 440, 1100, 2200\}$ entries for $f \in \{0.1, 0.2, 0.5, 1, 2, 5, 10, 20, 50, 100\}\%$; at $f = 100\%$ **every** gate is jittered.

Each perturbed init (3 seeds per f) is trained for 1008 steps of top-10% MSE on the full 500-image DIV2K-8q pool (mini-batch 50, a **fixed** shuffle seed so only the perturbation varies), then scored for test PSNR at four keep ratios ρ on the fixed canonical 50-image test set. $f = 0$ is the undisturbed exact-init reference (its untrained PSNR reproduces the classical DCT-IV: 30.54 dB @ $\rho\{=\}.20$).

Final PSNR vs. disturbance

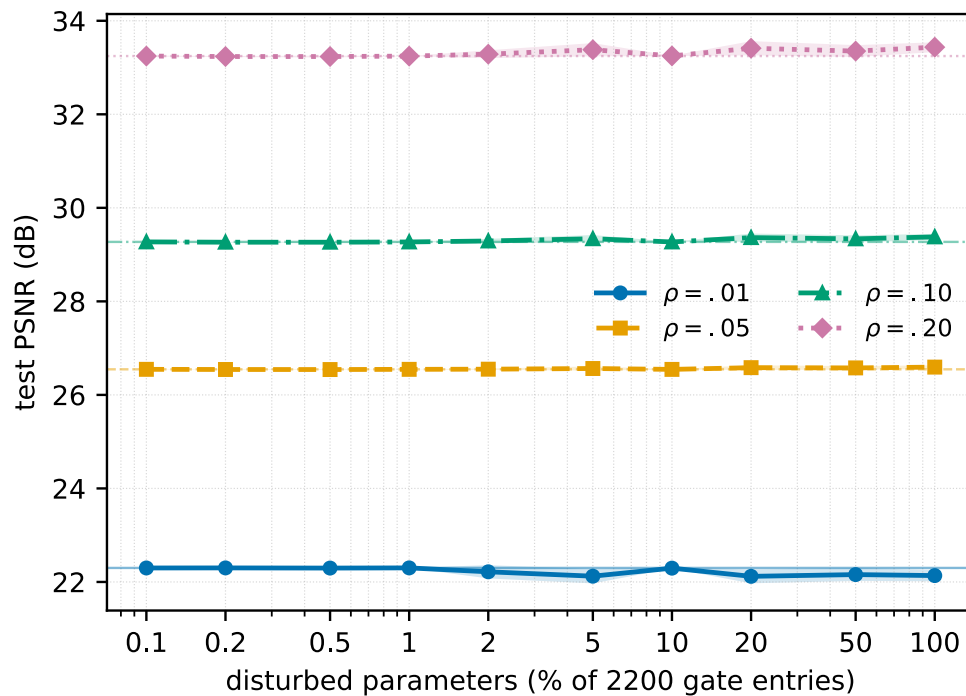


Figure 1: Trained test PSNR vs. the fraction of exact-init parameters disturbed (log-x), one curve per keep ratio ρ , mean $\pm \sigma$ over 3 seeds (shaded). Thin horizontal lines mark the undisturbed exact-init-trained PSNR per ρ .

disturbed	$\rho\{=\}.01$	$\rho\{=\}.05$	$\rho\{=\}.10$	$\rho\{=\}.20$
0 (exact)	22.3	26.55	29.27	33.25
0.1%	22.3 $\pm$ 0	26.55 $\pm$ 0	29.27 $\pm$ 0.01	33.25 $\pm$ 0.01
0.2%	22.3 $\pm$ 0	26.54 $\pm$ 0	29.27 $\pm$ 0.01	33.24 $\pm$ 0.01

disturbed	$\rho\{=\}.01$	$\rho\{=\}.05$	$\rho\{=\}.10$	$\rho\{=\}.20$
0.5%	22.3 ± 0	26.54 ± 0.01	29.26 ± 0.01	33.24 ± 0.02
1%	22.3 ± 0	26.55 ± 0	29.27 ± 0	33.24 ± 0
2%	22.22 ± 0.14	26.55 ± 0.01	29.29 ± 0.05	33.29 ± 0.08
5%	22.12 ± 0.15	26.56 ± 0.04	29.34 ± 0.08	33.38 ± 0.14
10%	22.3 ± 0.01	26.55 ± 0	29.27 ± 0	33.25 ± 0
20%	22.12 ± 0.15	26.58 ± 0.05	29.36 ± 0.09	33.41 ± 0.15
50%	22.16 ± 0.14	26.58 ± 0.04	29.34 ± 0.08	33.35 ± 0.13
100%	22.14 ± 0.12	26.59 ± 0.04	29.38 ± 0.07	33.44 ± 0.11

Trained test PSNR (dB), mean $\pm \sigma$ over 3 perturbation seeds; row 0 (**exact**) is the undisturbed exact-init-trained reference.

Recovery: perturbed init vs. trained

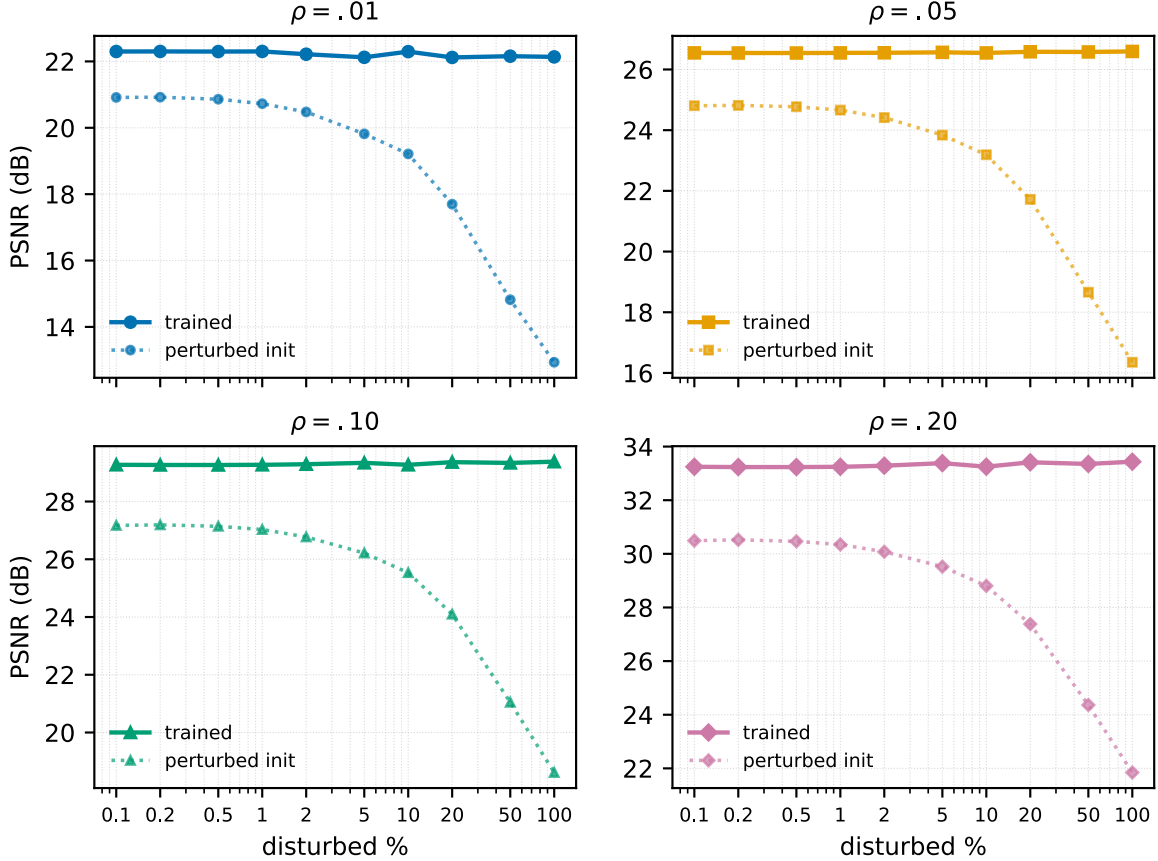


Figure 2: Per- ρ panels: the perturbed **init** PSNR (dotted) drops with f , while the **trained** PSNR (solid) stays close to the exact-init reference — the gap between the two curves is the amount training recovers.

Reading

At $\rho\{=\}.20$ the undisturbed exact init trains to 33.25 dB, and **no amount of disturbance moves that endpoint**: across the entire range — from 0.1% up to **100%** of the 2200 gate entries jittered ($\sigma = 0.1$), i.e. every gate perturbed — trained PSNR stays 33.2– 33.4 dB at $\rho\{=\}.20$ and is equally flat at $\rho\{=\}.01/.05/.10$, never leaving the undisturbed reference by more than ~ 0.2 dB (within seed scatter). The perturbed **init**, by contrast, collapses monotonically — from 30.49 dB at 0.1% to 21.85 dB at 100%, an 8.7 dB fall below the exact init (30.54 dB) that leaves the untrained operator barely above the $\rho\{=\}.20$ floor — yet 1008 steps of top-10% training recover it to 33.44 dB, the exact-init endpoint.

So the exact DCT-IV init sits in a **very wide, flat basin**: a per-gate jitter of $\sigma = 0.1$ applied to **all** its parameters still drains to the same trained optimum. What the trained model depends on is the DCT-IV **topology** plus the top- k objective, not the precise gate values — provided each gate is only locally perturbed. The lever that would break recovery is thus the jitter **magnitude** σ (fixed here), not the disturbed **fraction**: contrast the random-init seed study, where fully Haar-random gates (an $O(1)$, not $\sigma\{=}0.1$, per-gate displacement) settle into a lower basin below the exact transform.